OCEANOGRAPHY

→ Oceanography→ about water substances (majorly w.r.t oceans)

→ Oceans contain saline water and covers 71% of the earth's surface & about 97% of earth's water.

→ Water body has specific characteristic features viz., salinity, temperature, density.

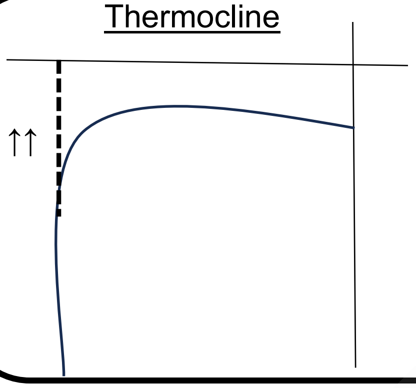
→ Water sources → Fresh (3%)
→ Marine (97%)

Water has high specific heat

→ Insolation determines temperature profile of an oceanic body.

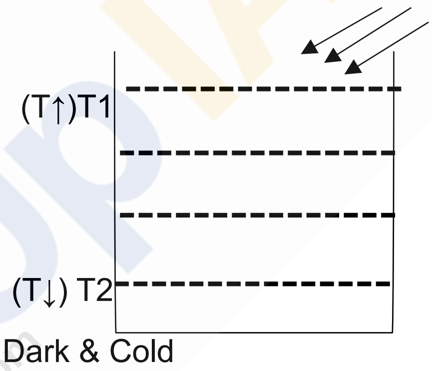
→ Evaporation takes places when the water body is subjected to temperature.

Thermocline

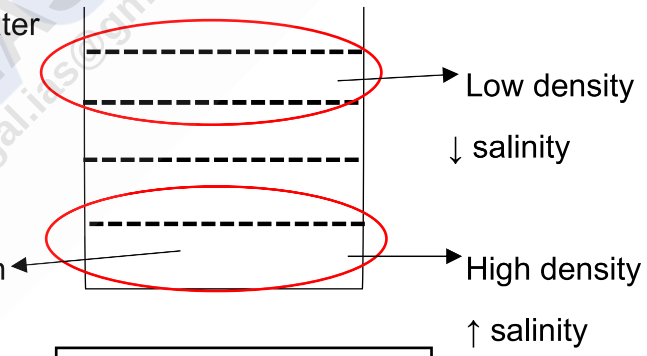


A point at which there is a sudden dip in the temperature of the water body.

Depth V/s temperature

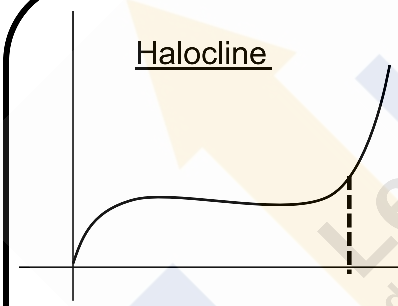


→ Density – Depends on the composition of water



Depth V/s salinity

Halocline



Sudden raise in the salinity of the water body.

→ with depth ------ (increasing): Salinity ↑, temperature ↓, density ↑

→ S T D relation → Densiity constatnt => T↑, S↑

Salinity constatant => T↑, D↓

Temperature constant => D↑, S↑

→ Sources of water

Oceans
Ice Caps and Glaciers
Groundwater
Lakes
Soil Moisture
Atmosphere
Streams and Rivers
Biosphere

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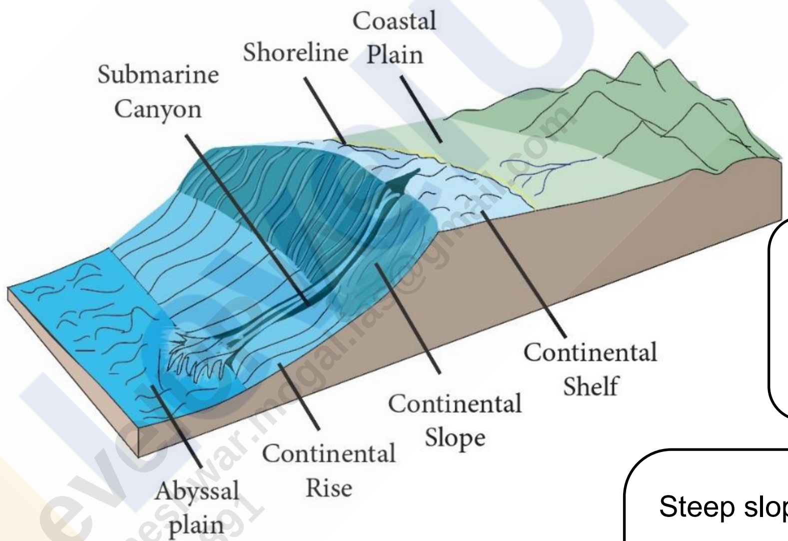
→ Water Cycle → The continuous circulation of water in the Earth – atmosphere system

→ Processes: Evaporation, transpiration, condensation, precipitation, and runoff

→ Oceanic Relief: it means kinds of landforms / topography in Ocean



Transitionary region
Less gradient
Less slopy

Steep slope
Most of the oceanic landforms are found (valleys, trenches)

Extension of continental rise

Oozes: Regions where pelagic & terragenous deposits present.

→ Pelagic : Bony structure (organic)

→ Terragenous: Muddy Deposits

Significance of continental shelf:

→ Rich fishing ground.

→ Presence of minerals & oils.

→ It supports marine ecosystem.

→ Helps shipping.

→ Tourism and marine culture.

→ Safeguards from natural calamities like cyclone (by growth of mangroves).

Deep sea plain / Abyssal plain:

- Starts beyond C.rise
- Formed by accumulation of sediments on seafloor.
- Many features like seamounts, guyots, atolls, sills and trenches are superimposed.

<u>Minor Oceanic Relief features:</u>

1. Mid Oceanic ridges :
   - found along diverging plate boundaries
   - Composed of two chains of mountains separated by a large depression
   - Mountains ranges have peaks as high as 2500m & even reach above surface.
2. Mountain structures : (i) Sea mount (ii) Guyots
   - (i) Sea mount
     - Undersea mountains
     - Pointed tips
     - Formed by volcanic activity
   - (ii) Guyots
     - flat topped seamounts.
     - Formed by the gradual sinking of sea floor
3. Trenches :
   - long, narrow, and deep depressions of the sea floor, with steep sides
   - Deepest parts of the oceans.
   - Associated with active volcanoes and earthquakes.
4. Atolls:
   - Ring-shaped coral reef, island, or series of islets
   - Surrounds a body of water called a lagoon.
   - Atolls develop with underwater volcanoes, called seamounts.

Bank, shoal, reef ------- explained in class

5. Submarine canyons:
   - Deep valleys beginning at edge of C.shelf
   - Largest canyon in world – Bering sea of Alaska.

Bays, gulfs, straits, Isthmus ------- explained in class

<u>Vertical & Horizontal distribution of temperature & salinity:</u>

Diurnal range of ocean temperatures is small because the process of heating & cooling of oceanic water is slower than land.

Factors affecting temperature:

- Insolation
- Heat loss due to reflections, scattering, evaporation & radiation.
- Albedo.
- Physical characteristic features of sea.
- Presence of submarine ridges.
- Shape of ocean.
- Enclosed sea in lower latitudes – high temperature than open seas & vice versa for higher latitudes.
- Lower weather conditions.
- Unequal distribution of land & water.
- On shore & off shore winds
- Ocean currents

Factors affecting Ocean salinity:

- Depends on evaporation & precipitation.
- Influenced by fresh water flow from rivers.
- Winds
- Ocean currents.

Salinity of Arabian sea > Bay of Bengal
Baltic sea- lower salinity
Mediterranean sea – high salinity
Black sea – very low salinity
Land locked sea – high salinity
Hot regions – high salinity
Pacific ocean- larger area- ↑ salinity
Atlantic ocean – Max. salinity (gradually decreases towards north)

Concept of ocean currents --- explained in class

Tides:

- Periodic raise & fall of seal level once (or) twice a day.
- Due to gravitational pull of sun & moon.
- Occurs due to balance of forces like centrifugal & gravitational.
- **Tidal Bulge:** Elongation of earth's oceans by gravitational force of moon.

Factors controlling tides:

- Movement of the moon
- Position change in sun & moon
- Uneven distribution of water.
- Irregularities in configuration of oceans.
- Shape of the bays & estuaries along coast line.

Types of tides

- Based on sun, moon & earth's position
  - Spring tides
  - Neap tides
- Based on frequency
  - Semi Diurnal tide
  - Diurnal tide
  - Mixed tide

<u>Characteristics of tides:</u>

- Tidal bulges – wide on C.shelf have greater height.
- Tidal currents in open oceans – weak.
- Shape of bay & estuary – magnify intensity of tides.
- Funnel shaped bays – change tidal magnitudes.
- Tidal patterns vary – across different oceans & locations.
- **Tidal bore** – the steep nosed tide crest looks like a vertical wall of water rushing upstream (kind of tidal current).

Characteristics of tidal bore:
- less predictable & dangerous.
- Affect shipping & navigation.
- Disrupt fishing zones.

Importance of tides:
- High tides- favourable for navigation.
- Higher tides- enables bigger catch.
- Desilts the sediments & removes polluted water.
- Enables renewable form of energy.